research_decisions.md

Project: Comparative Analysis of AI-Assisted and Manual Data Labeling Accuracy Across Scientific Datasets **Goal:** Convert a completed literature review into an implementation-based research paper **Log opened:** 2026-10-21 | **Current stage:** 0 (Intake) -> 1 (Gap identification)

0. Conventions

Evidence tags (applied to every substantive statement):

- **VERIFIED** - confirmed in a file, source, or output that has been shown and read.
- **INFERRED** - a reasoned conclusion from verified material; the reasoning is stated.
- **UNKNOWN** - no evidence yet; the file, output, or check that would resolve it is named.

Decision status: ADOPTED (settled in the project brief) | PROPOSED (needs confirmation) | OPEN (needs evidence before deciding).

Entry template: ID - Decision - Reason - Source/Evidence (with tag) - Next action.

Standing rules: no invented papers, datasets, repository links, code execution, experiments, or metrics. Every number in the final manuscript must trace to an output that was actually produced and shown.

1. Source review: scope as read from the attached PDF

Everything below was read from the attached file only. Nothing outside it has been checked.

#	Item	Tag
S1	IEEE-style two-column paper: Abstract, 8 numbered sections (I-VIII), 3 tables, 2 figures, 3 numbered equations, 79 references.	VERIFIED
S2	Type: narrative review. No search protocol, databases, date range, inclusion/exclusion criteria, or screening process is stated. Whether one was followed is not visible in the file.	VERIFIED (absent from text); protocol existence UNKNOWN
S3	No research questions, hypotheses, or experiments are stated.	VERIFIED
S4	Section map: I Introduction; II Theoretical and statistical frameworks for ground truth (annotation as measurement, IAA metrics); III Medical imaging and computational pathology (SAM/MedSAM, annotation throughput, HybridMS); IV Genomics and clinical text (scRNA-seq LLM annotation, clinical trial data cleaning); V Astrophysics and geospatial remote sensing (galaxy morphology only); VI Psychology and ergonomics (fatigue, automation bias/anchoring); VII Label-noise mitigation (active learning, confident learning); VIII Conclusion.	VERIFIED
S5	Domains in the abstract: computational pathology, single-cell transcriptomics, clinical trial management, geospatial remote sensing, astrophysics.	VERIFIED
S6	Domains with body content: medical imaging and pathology, scRNA-seq, clinical trial data cleaning, galaxy morphology. Geospatial remote sensing has no body subsection.	VERIFIED
S7	Central thesis in the conclusion: AI assistance speeds labeling and reduces fatigue and inter-observer variability, but adds automation bias and anchoring risk; best strategy is stated to rely on active learning and confident learning.	VERIFIED (what the text says); strength of support see E6
S8	Metrics defined: Cohen's kappa, Fleiss' kappa, Krippendorff's alpha, STAPLE, Dice (DSC), IoU.	VERIFIED
S9	Source authorship/provenance: the title page lists one author. Whether this is the researcher's own work, co-authored work, or a shared draft is not stated in the file.	UNKNOWN - confirm (see D-004)

2. Evidence audit

2.1 Findings

ID	Finding	Where	Tag	Action
E1	**Citation-claim mismatches inside the document.** (a) Sec. III-B cites [7] (brain tumour segmentation benchmarking) as "a benchmark study" of annotation peripherals and cites [1] (statistical framework for human labeling) for the 29.9-minute / 24% mouse-annotation result; the reference list shows the pathology pilot study as [9]. (b) Sec. VII-A cites ActiveAED as [23], but [23] in the list is a conformal-in-the-loop paper; ActiveAED is [71]. (c) Active-learning principle is cited to [21], which is a retinal-image confident-learning paper.	III-B, VII-A, refs	VERIFIED (mismatch between text and list); correct source UNKNOWN	Re-map each claim to its true source after opening the source
E2	**Uncited references.** I found no in-text citation for [24]-[79] (56 of 79 entries). This includes the geospatial references [51]-[56] and the automation-bias-in-pathology paper [63].	Sections I-VIII	VERIFIED (from the text provided)	Decide per reference: cite, or remove
E3	**Uncited quantitative claims.** The oral-lesion kappa = 0.90, the "substantial label noise in curated benchmarks" claim (abstract), and the VLM/VQA transition statement (Sec. V-A) carry no citation.	Abstract, II-B, V-A	VERIFIED	Source each or delete

E4	**Reference list quality.** Most entries use placeholder authors ("Author Team N", "A. T. N") instead of real author names. Titles often include site suffixes (e.g., "pdf", "researchgate", "mdpi", "scribd"). Duplicates: [13], [43], [44] share one title and arXiv ID; [14] and [48] share one title and arXiv ID. Some URLs look malformed: [3] has non-hexadecimal characters in what looks like a UUID; [53] has a broken GitHub path; [55] has a truncated DOI; [63] has an arXiv ID of unusual shape.	References	VERIFIED (as printed); whether URLs resolve UNKNOWN	Rebuild the list from the real source pages
E5	**Citation years.** Every entry is dated 2026, but several arXiv IDs encode earlier months (e.g., 2408.xxxxx, 2504.xxxxx, 2508.xxxxx, 2503.xxxxx). The year appears to be access year, not publication year.	References	INFERRED (arXiv IDs encode YYMM)	Replace with true publication years/venues
E6	**Weak sources behind key psychology claims.** The 30% fatigue error rate cites a vendor blog [18]; fatigue [17] is a company blog; "Weight of Advice" [20] is a document-hosting upload; several IAA definitions rest on vendor or aggregator pages ([3], [26], [28]).	VI, II	VERIFIED (sources as listed); weakness INFERRED	Replace with primary or peer-reviewed sources
E7	**Construct mismatch with the title.** The Dice values in Table II mostly compare model outputs (zero-shot SAM, MedSAM, nnU-Net) against reference masks. That is model accuracy, not accuracy of an AI-assisted human workflow versus a manual workflow. Workflow-level comparisons in the text are [9] (time, variability), [10] (time, Likert grade), [11] (Dice, time) and [13] (error rate, throughput), and they use different outcome measures.	III, IV, Table II	INFERRED from the text	Define "labeling accuracy" before choosing experiments (see D-007)
E8	**Table II cells without support.** "Efficiency gain: High" for SAM and MedSAM and "None" for nnU-Net have no numbers in the text. The nnU-Net row says "Lumbar/Glioma, DSC 0.99", but the text gives 0.99 for lumbar structures only; the glioma passage reports 0.894 as an upper bound for SAM with precise boxes.	Table II, III-A	VERIFIED (text vs table)	Correct or remove
E9	**Overreach in the conclusion.** (a) "Decrease inter-observer variability" rests on one pilot study [9]. (b) "Successfully mitigate annotation fatigue" - no fatigue measurement across sessions is reported. (c) "Replacing subjective manual literature reviews" appears to refer to manual cell-type annotation; the wording does not match the body. (d) "Optimal strategy relies heavily on active and confident learning" - the body presents no comparison against alternatives. (e) Sec. III-B explains higher AI-assisted Likert grades as AI uncovering boundary details that fatigue hides - a causal explanation not tested in the text.	III-B, VIII	INFERRED	Soften wording or add evidence
E10	**Imprecise technical statement.** Table I describes Dice/IoU as "harmonic mean of precision and recall". By the review's own equations (2)-(3), that holds for Dice only, not IoU.	Table I	VERIFIED (against eqs. 2-3)	Fix
E11	**Small effects reported without uncertainty.** HybridMS vs MedSAM: Dice 0.9538 vs 0.9435 (difference 0.0103). GB-SAM vs U-Net: 0.885 vs 0.857 (difference 0.028). No confidence intervals, test statistics, or seed counts appear in the review. Whether the sources report them is unknown.	III-A, III-C	VERIFIED (differences computed from review values); source reporting UNKNOWN	Check sources; plan uncertainty reporting in own experiment
E12	**Domain imbalance.** Abstract promises geospatial remote sensing; body has none. Genomics body = scRNA-seq only. Astronomy figures (87%, about 97%) are model-vs-expert-label results, not human-vs-AI-assisted comparisons.	Abstract, IV, V	VERIFIED / INFERRED	Narrow or rebalance scope
E13	**Figures 1 and 2** contain small numeric tables (time effort, kappa, Dice, query counts) with no source in the captions. The values cannot be read reliably at the resolution provided. They may be illustrative.	Figs. 1-2	UNKNOWN	Provide figure source files or confirm they are illustrative
E14	**Structure.** The abstract stops after defining the HITL paradigm and states no findings. The introduction begins mid-thought, continuing the abstract's last sentence.	Abstract, I	VERIFIED	Fix when the manuscript is rebuilt
E15	**Arithmetic spot-checks.** Octozi: 54.67 / 8.48 = 6.447, reported as 6.44 (truncated rather than rounded; check the source). Pathology: 29.9 -> 13.6 minutes is a 54.5% reduction.	IV-B, III-B	VERIFIED (computed)	None beyond source check

2.2 Claim register (numbers to verify against source files)

All entries are **UNKNOWN** until the cited source has been opened and the number located in it.

ID	Claim as written in the review	Cited as
C1	Fleiss' kappa 0.383 (internal) and 0.255 (external), eleven ICU consultants	[2]
C2	Oral potentially malignant disorders, expert kappa = 0.90	uncited
C3	MedSAM training set: 1,570,263 image-mask pairs, 10 modalities, 30+ cancer types	[5]
C4	Lumbar MRI: SAM Dice 0.64 (discs), 0.83 (vertebrae); MedSAM 0.79, 0.88; nnU-Net 0.99	[6]
C5	BraTS 2023: SAM/SAM 2 up to 0.894 Dice with precise box prompts	[7]
C6	CRAG: GB-SAM 0.885 with 25% of data vs U-Net 0.857	[8]
C7	WSI annotation: mouse 29.9 min / 24%; touchpad 47.5 min / 45%; SAM-assisted 13.6 min / 2%	[1] and [9] (see E1)
C8	AICL: 2.8x less effort than clinical staff editing, 8.7x less than expert from scratch; Likert 8.4 vs 7.8	[10]
C9	HybridMS: Dice 0.9538 vs MedSAM 0.9435; time down 82% (60% on challenging cases)	[11]
C10	LLM cell-typing benchmark across 34 datasets; outperforms SingleR, Cellmarker2.0, ScType	[12]
C11	Octozi: throughput 6.03x; cleaning errors 54.67% -> 8.48%	[13]
C12	Zoobot lopsided-galaxy accuracy 87%; YOLOv5 edge-on detection about 97%	[15], [16]

C12	200bot lopsided-galaxy accuracy 87%; YOLOv5 edge-on detection about 97%	[15], [16]
C13	Prolonged manual annotation: error rates up to 30%	[18]
C14	Active learning reduces annotation budget by up to 99%	[22]

3. Decision log

Adopted (from the project brief)

D-001 - Evidence-tagging protocol

- Decision: every substantive statement is tagged VERIFIED, INFERRED, or UNKNOWN.
- Reason: keeps claims traceable and prevents unverified material from entering the manuscript.
- Source: project brief. Status: ADOPTED.
- Next action: apply to all later stages.

D-002 - Staged workflow

- Decision: gap identification -> literature-to-experiment mapping -> feasibility -> research questions -> verified baseline -> coding -> experimental evaluation -> manuscript and reproducibility.
- Reason: each stage produces evidence the next stage depends on.
- Source: project brief. Status: ADOPTED.
- Next action: proceed to Stage 1.

D-003 - No fabrication

- Decision: no invented papers, datasets, repository links, code execution, experiments, or metrics. Any result in the paper must come from an output that was run and shown.
- Reason: research integrity and reproducibility.
- Source: project brief. Status: ADOPTED.
- Next action: request source files or outputs whenever evidence is missing.

Proposed (need confirmation)

D-004 - Confirm provenance of the source review

- Decision: record who authored the review and how the extension paper will credit it.
- Reason: the title page lists one author (S9); authorship and reuse permission affect how the new paper is framed and submitted.
- Source: S9 (UNKNOWN). Status: PROPOSED.
- Next action: researcher confirms ownership/co-authorship and any supervisor or course requirements.

D-005 - Rebuild the reference list before any reuse

- Decision: do not carry the existing reference list into the new manuscript. Rebuild it from the actual source pages with real authors, venues, years, and working links; keep only references that are cited and verified.
- Reason: E1-E6 (mismatched citations, placeholders, duplicates, uncited entries, weak sources, wrong years).
- Source: E1-E6 (VERIFIED as printed). Status: PROPOSED.
- Next action: verify load-bearing sources first: [2], [5], [6], [9], [10], [11], [13], [63].

D-006 - Describe the review honestly as a narrative review

- Decision: the extension paper presents the review as a narrative synthesis and does not claim systematic-review methodology.
- Reason: no search protocol is stated (S2).
- Source: S2. Status: PROPOSED.
- Next action: if a search protocol exists, provide it; otherwise adopt this wording.

D-007 - Define "labeling accuracy" before designing experiments

- Decision: operationalize accuracy as agreement between the final labels from each workflow (manual vs AI-assisted) and an adjudicated reference, reported with uncertainty, alongside time and annotator-agreement outcomes.
- Reason: the review mostly reports model-vs-reference accuracy (E7), which does not answer the title's question; outcome measures differ across studies (Dice, error %, Likert, kappa, minutes).
- Source: E7 (INFERRED). Status: PROPOSED.
- Next action: confirm in Stage 1; then check which public datasets provide both adjudicated references and multi-annotator labels (not yet identified; UNKNOWN).

Open (need evidence)

ID	Question	Evidence needed
O-1	Primary research question and hypotheses	Stage 1 gap analysis; supervisor expectations
O-2	Which domain(s) and dataset(s) to use (the review spans four; one or two is likely feasible)	Compute and time limits; dataset licences and label availability
O-3	Source of "manual" labels: existing multi-annotator labels, or new annotation	Dataset documentation; whether human annotators can be recruited
O-4	Is a new experiment required, or is a smaller replication acceptable	Course brief or supervisor
O-5	Compute environment (CPU/GPU, Colab, local, cluster)	Researcher input
O-	Deadline, venue, format, length	Researcher input / rubric

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4. Provisional gap candidates (INFERRED - to be tested in Stage 1)

ID	Candidate gap	Basis
G1	Few studies in the review directly compare AI-assisted and manual workflows against a common reference with accuracy as the primary outcome.	E7
G2	No common accuracy scale across domains; outcomes are Dice, error rate, Likert grade, kappa, and time.	E7, C-register
G3	Automation-bias and anchoring evidence relies on weak sources; the one computational-pathology study on it ([63]) is listed but never cited.	E2, E6
G4	Reported gains are small in places and carry no stated uncertainty.	E11

These are candidates only. None becomes a research gap until the supporting sources are opened and read.

5. Evidence requests

1. Confirmation of authorship and provenance of the source review (D-004).
2. PDFs, or permission to fetch and read, the load-bearing sources: [2], [5], [6], [9], [10], [11], [13], [63].
3. Source files or data behind Figures 1 and 2, or confirmation that the values are illustrative (E13).
4. Deadline, venue or rubric, expected length (O-6).
5. Compute environment (O-5).
6. Any datasets, labeled samples, annotation tools, or code already in hand.
7. Whether a new experiment is required or a replication is enough (O-4).

6. Next actions

#	Action	Stage	Depends on
N1	Confirm D-004 to D-007 or amend them	0	Researcher
N2	Open the load-bearing sources and mark each claim C1-C14 VERIFIED or not	1	Evidence request 2
N3	Turn gap candidates G1-G4 into evidence-backed gaps	1	N2
N4	Map each confirmed gap to a testable experiment	2	N3
N5	Feasibility check against data, compute, time	3	O-2, O-5, O-6
N6	Draft research questions and hypotheses	4	N4, N5

Log entries are appended as decisions are made. Superseded entries are marked, not deleted.